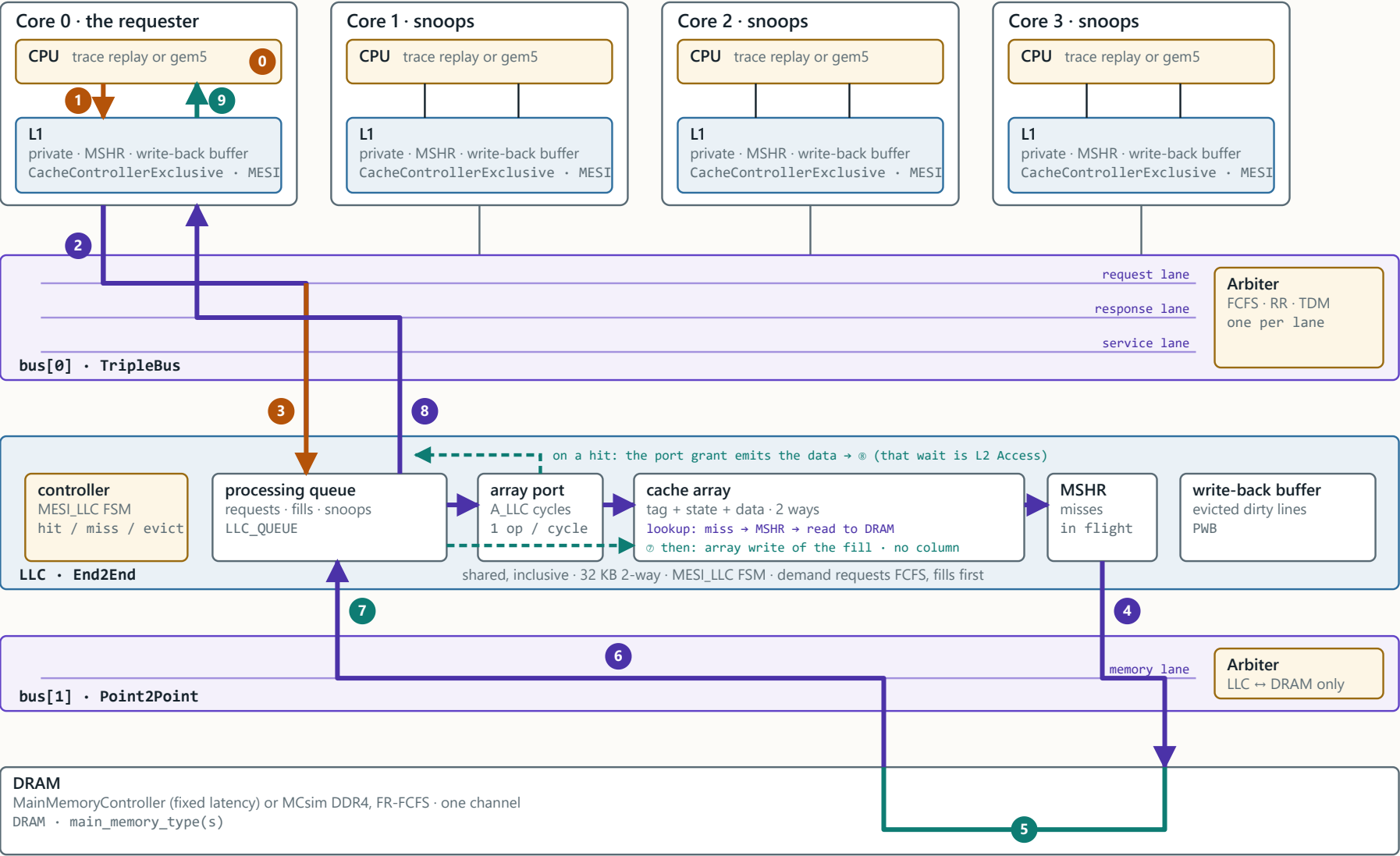


The system, end to end — one LLC miss from core 0, and the column that measures each hop

MultiCoreSystem as shipped: 4 cores, snoop MESI on a TripleBus, one shared LLC, a point-to-point memory bus, DRAM. Colour = the LatencyReport column; ①–⑨ are the report's nine latency columns in order, at the Logger stamps that bound them.



The route, hop by hop

stamp → stamp is the column's span

- ① Ready; waits for a slot in the OoO window
Ready Cycle → CPU. ENTER
CPU Latency
- ② Issued; queued at L1
CPU. ENTER → L1. ENTER
L1 Stall
- ③ L1 misses, MSHR; wins the request lane
L1. ENTER → REQ_BUS. EXIT
Request Bus
- ④ Waits in the LLC queue; every L1 snooped it
REQ_BUS. EXIT → LLC. ENTER
L2 Stall
- ⑤ Lookup misses; MSHR; read leaves on bus[1]
LLC. ENTER → DRAM. ENTER
L2-DRAM Bus
- ⑥ DRAM serves the read
DRAM. ENTER → DRAM. EXIT
DRAM
- ⑦ Data back on bus[1], to the queue front
DRAM. EXIT → MEM_BUS. EXIT #2
L2-DRAM Bus — the same column as ④
- ⑧ Fill processed; response emitted
MEM_BUS. EXIT #2 → LLC. SERVICE=EXIT
L2 Access ~1 cycle; array write off-path
- ⑨ Response lane back to core 0
LLC. EXIT → RESP_BUS. EXIT
Response Bus
- ⑩ L1 fills, CPU retires it
RESP_BUS. EXIT → CPU. EXIT
L1 Access

Total = delivered – issued = ① ... ⑨, tiled exactly; ① lies before it.
Under MCsim there are no DRAM stamps: ④ runs to MEM_BUS.EXIT #1, ⑤ is the whole bus[1] round trip, ⑥ is 0.

- The other cores their L1s see every request on the request lane; an owner answers on the response lane in the LLC's place, its stamps standing in for the LLC's in ③ ④.
- The service lane the LLC's back-invalidations on an eviction and their echoes; not on this request's path.
- Directory protocol the same boxes: requests become point-to-point and the LLC-side directory forwards them instead of the L1s snooping.

waiting: in a queue, for a lane, for the port moving: a bus transfer being served: an array or DRAM access off the critical path